

Boards quantity

1

Build All

Export All

Export

Build BOM

Legend

Item #	Quantity	Part Ref	Value	Description	Mfg	Mfg Part Number	Availability	
							DigiKey	Mouser Electronics
1	1	BR1	Z4GP208L-HF	800 V, 2 A, Standard Recovery Bridge, 4-SMD	Comchip Technology	Z4GP208L-HF		
2	1	C1	47 nF	47 nF, 250 V, Film, X Class	Murata	GA355ER7GB473KW01L		
3	1	C2	120 µF	120 µF, 400 V, High Voltage Al Electrolytic, (35.5 mm x 18 mm)	Nichicon	UPT2G121MHD6		
4	1	C3	0.82 nF	0.82 nF, 630 V, High Voltage Ceramic	Murata	GRM31B5C2J821JW01L		
5	2	C4, C6	4.7 µF	4.7 µF, 10.0 V, Ceramic, X7R	TDK	C1608X7S1A475K080AC		
6	2	C5, C13	2.2 µF	2.2 µF, 16 V, Ceramic, X7R	TDK	CGA4J3X7R1C225K125AB		
7	1	C7	0.68 nF	0.68 nF, 250 VAC, Ceramic, Y Class	Murata	GA352QR7GF681KW01L		
8	1	C8	820 pF	820 pF, 200 V, High Voltage Ceramic	AVX Corp	08052C821KAT2A		
9	1	C9	22 µF	22 µF, 100 V, Electrolytic, Low ESR, 1300 mΩ, (10.2 mm x 8 mm)	Panasonic	EEE-TG2A220UP		
10	2	C10, C11	680 µF	680 µF, 35 V, Al Organic Polymer, 11 mΩ, (16.5 mm x 10 mm)	Panasonic	EEH-ZS1V681UP		
11	1	C12	1 µF	1 µF, 50 V, Ceramic, X7R	Kemet	C1206C105K5RACTU		
12	1	D1	RS07K-GS08	800 V, 1.4 A, Fast Recovery, 300 ns,	Vishay	RS07K-GS08		

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				DO-219AB				
13	1	D2	BAV21WS-7-F	250 V, 0.25 A, Standard Recovery, SOD-323	Diodes Inc.	BAV21WS-7-F	✓	✓
14	1	D3	SM4001PL-TP	50 V, 1 A, Standard Recovery, SOD-123FL	MCC	SM4001PL-TP	✓	✓
15	1	F1	1.25 A	350 VAC, 1.25 A, Glass Cartridge, Time Lag Fuse	Bel Fuse Inc.	2JS 1.25-R	✓	✓
16	1	L1	6 mH	6 mH, 1.6 A	Panasonic	ELF18N016	⚠	⚠
17	1	M1	AOSP66923	MOSFET, N- Channel, 100 V, 9.5 A, SOIC-8	Alpha & Omega Semiconductor Inc.	AOSP66923	✓	⚠
18	1	M2	AOD66406	MOSFET, N- Channel, 40 V, 45 A, TO-252 (DPAK)	Alpha & Omega Semiconductor Inc.	AOD66406	✓	⚠
19	1	Q1	MMBTA06LT1G	NPN, Small Signal BJT, 80 V, 500 mA, SOT-23-3	ON Semiconductor	MMBTA06LT1G	✓	✓
20	2	R1, R2	300 kΩ	300 kΩ, 5 %, 0.5 W, Thick Film	Generic		⚠	⚠
21	1	R3	6.8 Ω	6.8 Ω, 5 %, 0.125 W, Thick Film	Generic		⚠	⚠
22	1	R4	182 kΩ	182 kΩ, 1 %, 0.125 W, Thick Film	Generic		⚠	⚠
23	1	R5	1.33 kΩ	1.33 kΩ, 1 %, 0.125 W, Thick Film	Generic		⚠	⚠
24	1	R6	47 Ω	47 Ω, 5 %, 0.125 W, Thick Film	Generic		⚠	⚠
25	2	R7, R8	1.78 MΩ	1.78 MΩ, 1 %, 0.25 W, Thick Film	Generic		⚠	⚠
26	1	R9	8.25 mΩ	8.25 mΩ, 1 %, 0.25 W, Metal Film	Generic		⚠	⚠

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27	1	R10	324 kΩ	324 kΩ, 1 %, 0.125 W, Thick Film	Generic			
28	1	R11	10 Ω	10 Ω, 1 %, 0.125 W, Thick Film	Generic			
29	1	R12	11 Ω	11 Ω, 5 %, 0.5 W, Thick Film	Generic			
30	1	R13	100 Ω	100 Ω, 5 %, 0.25 W, Thick Film	Generic			
31	1	RT1	4 Ω	NTC Thermistor 4 Ω, 5 A	TDK	B57211P0409M301		
32	1	T1	E30 (E30/15/7-3F3)	3F3 Core Material Refer to Manufacturer datasheet for a number of parts to purchase	Ferroxcube	E30/15/7-3F3		
33	1	T1 Bobbin	E30/15/7 - 1 (P7-S7)	Bobbin Material : GFR polyterephthalate	EPCOS (TDK)	B66232		
34	1	T1 Core Acc.1	B66232	Yoke . Stainless spring steel (0.4mm)	EPCOS (TDK)	B66232		
35	1	T1 Core Acc.2	CLA-E30/15/7	Clasp . CuZn Alloy, Ni plated	Ferroxcube	CLA-E30/15/7		
36	1	T1 Core Acc.3	SPR-E30/15/7	Spring . Stainless steel (CrNi)	Ferroxcube	SPR-E30/15/7		
37	1	U1	INN3370C-H302	InnoSwitch3-Pro, INN3370C-H302, inSOP-24D	Power Integrations	INN3370C-H302		
38	1	VR1	TZMB9V1-GS08	9.1 V, 500 mW, 2 %, SOD-80, General Purpose	Vishay	TZMB9V1-GS08		
39	1			104 mm² area on Copper PCB. 2 oz (70 μm) thickness.	Custom			

Item #	Quantity	Part Ref	Value	Description	Mfg	Mfg Part Number	Availability	
							DigiKey	Mouser Electronics
				Heatsink for use with Device U1.				
40	1			192 mm ² area on Copper PCB. 2 oz (70 μm) thickness. Heatsink for use with Rectifier M1.	Custom		✓	✓